```cpp
#include <iostream>
using namespace std;
int main() {
    int n, k;
    cout << "Enter n and k: ";
    cin >> n >> k;
    int C[50][50];
    for (int i = 0; i <= n; i++) {
        cout << "\nRow " << i << ":\n";
        for (int j = 0; j <= i; j++) {
            if (j == 0 || j == i) {
                C[i][j] = 1;
                cout << "  C(" << i << "," << j << ") = 1\n";
            } else {
                C[i][j] = C[i - 1][j - 1] + C[i - 1][j];
                cout << "  C(" << i << "," << j << ") = C(" << i - 1 << "," << j - 1 << ") + C(" << i - 1
                        << "," << j << ") = "
                     << C[i - 1][j - 1] << " + " << C[i - 1][j] << " = " << C[i][j] << endl;
            }
        }
        cout << "  Row " << i << ": ";
        for (int j = 0; j <= i; j++)
            cout << C[i][j] << " ";
        cout << endl;
    }
    cout << "\nC(" << n << "," << k << ") = " << C[n][k] << endl;
    return 0;
}
```

```
PS C:\Users\fishb\Desktop\Sem 7\DAA> cd "c:\Users\fishb\Desktop\Sem 7\DAA\" ; if ($?) { g++ 5_binomial_coefficient.cpp -o
 5_binomial_coefficient } ; if ($?) { .\5_binomial_coefficient }
Enter n and k: 5
3

Row 0:
  C(0,0) = 1
  Row 0: 1

Row 1:
  C(1,0) = 1
  C(1,1) = 1
  Row 1: 1 1

Row 2:
  C(2,0) = 1
  C(2,1) = C(1,0) + C(1,1) = 1 + 1 = 2
  C(2,2) = 1
  Row 2: 1 2 1

Row 3:
  C(3,0) = 1
  C(3,1) = C(2,0) + C(2,1) = 1 + 2 = 3
```

```
Row 3:
  C(3,0) = 1
  C(3,1) = C(2,0) + C(2,1) = 1 + 2 = 3
  C(3,2) = C(2,1) + C(2,2) = 2 + 1 = 3
  C(3,3) = 1
  Row 3: 1 3 3 1

Row 4:
  C(4,0) = 1
  C(4,1) = C(3,0) + C(3,1) = 1 + 3 = 4
  C(4,2) = C(3,1) + C(3,2) = 3 + 3 = 6
  C(4,3) = C(3,2) + C(3,3) = 3 + 1 = 4
  C(4,4) = 1
  Row 4: 1 4 6 4 1

Row 5:
  C(5,0) = 1
  C(5,1) = C(4,0) + C(4,1) = 1 + 4 = 5
  C(5,2) = C(4,1) + C(4,2) = 4 + 6 = 10
  C(5,3) = C(4,2) + C(4,3) = 6 + 4 = 10
  C(5,4) = C(4,3) + C(4,4) = 4 + 1 = 5
  C(5,5) = 1
  Row 5: 1 5 10 10 5 1

C(5,3) = 10
PS C:\Users\fishb\Desktop\Sem 7\DAA>
```